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# Microstructure and mechanical properties of a Fe–28Mn–9Al-1.2C-(0, 3, 6, 9)Cr austenitic low-density steel

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ABSTRACT

In this work, the effects of Cr addition on microstructure, carbide formation and mechanical properties of Fe–28Mn–9Al-1.2C-*x*Cr ($x$ = 0, 3, 6, 9) strips fabricated under a near-rapid solidification condition are investigated. It is found that Cr addition makes as-cast strips present a nonlinear evolution on yield strength, i.e. firstly reduces and then increases beyond that of Cr-free strip, and the total elongation of as-cast strips clearly decreases. The evolution is related to precipitation reduction of κ-carbides, increased dislocation density, and formation of $Cr_{23}C_6$ instead of κ-carbides in 9Cr strip with Cr addition. Annealing at 550 °C for 5 h enhances yield strengths of all strips due to more carbide precipitations. An observed unusual phenomenon is that after annealing, Cr-free strip shows lower total elongation than the as-cast strip, while all Cr-containing strips show higher total elongation due to reduction of dislocations. Formation of $Cr_{23}C_6$ carbides at austenite grain boundaries is a reason for the lowest ductility of as-cast and annealed strips containing 9 wt% Cr. Analysis in 0Cr and 6Cr strips indicates that both slip bands and dislocations contribute to ductility of this austenitic low-density steel.

## 1. Introduction

To save energy and decrease $CO_2$ gas emission, automotive weight reduction has become an increasingly significant topic in recent years. Due to high specific strengths, Fe–Mn–Al–C low-density steels have attracted much attention from both academic and industrial fields [1–3]. It is known that austenitic low-density steels have good ductility, while their strengths are typically low [4–6]. Precipitation of κ-carbides can effectively improve the strengths of austenitic low-density steels [7–11]. However, the formation of coarsened κ-carbides would lead to a significant reduction in ductility [12–16]. Thus, regulating of carbide precipitation plays an important role in comprehensive performance of austenitic low-density steels.

It was found that Cr addition could affect the precipitation of κ-carbides and strain hardening rate of Fe–20Mn–9Al-1.2C low-density steels; as a result, improving strength and without loss of ductility [17]. Yuji et al. [18] also found that 5 wt% Cr addition inhibited formation of coarse κ-carbides in Fe–20Mn-*x*Al-*y*C low-density steel, which improved ductility. A study by Chen et al. [19] indicated that the microstructure of Fe–9Al–30Mn-0.6C-*x*Cr low-density steels transformed from α + B2 to α + DO3, when Cr content increased from 3.0 wt% to 7.0 wt%. Some researchers considered that Cr element could improve crack resistance of low-density steels through enhancing an inter-crystalline binding force [19–22]. Moreover, it was reported that Cr addition could obviously improve corrosion resistance, high temperature oxidation resistance and environmental brittleness of Fe–Mn–Al–C low-density steels [23,24].

Austenite stabilization effect of Mn and C element ensures ductility, solid solution strengthening of C element, and precipitation strengthening of κ-carbides, so high Mn and high C contents are essential to maintain a high-performance of Fe–Mn–Al–C austenite low-density steels [2,13,14]. However, effect of Cr element on microstructures and properties of Fe–Mn–Al–C low-density steels with high Mn and high C have received limited attention. Thus, this work systematically studied the effect of Cr in 0–9 wt% range on microstructure and mechanical properties of Fe–28Mn–9Al-1.2C low-density steels.

Previous work indicated that a near-net shape method prepared under near-rapid solidification condition was a simple and highly

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**Table 1**
Measured compositions of the studied steel strips in this work (wt. %).

| Strips | Fe | Mn | Al | C | Cr |
|---|---|---|---|---|---|
| 0Cr | Bal. | 26.6 | 9.1 | 1.2 | – |
| 3Cr | Bal. | 27.6 | 8.4 | 1.3 | 2.8 |
| 6Cr | Bal. | 29.1 | 9.9 | 1.2 | 6.8 |
| 9Cr | Bal. | 27.7 | 9.5 | 1.3 | 9.3 |

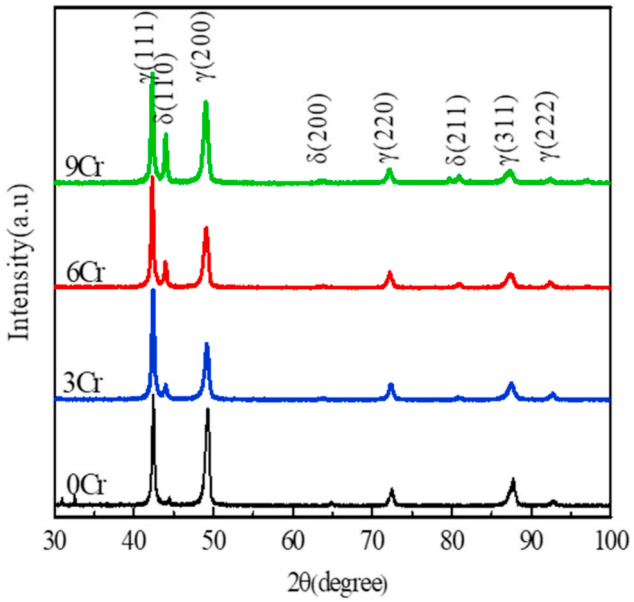



**Fig. 1.** X-ray diffraction patterns of as-cast Fe–28Mn–9Al-1.2C-xCr strips.

efficient method to manufacture high-strength low-density steel, which could avoid the problems such as coarse carbides and composition segregation of austenite Fe–Mn–Al–C low-density steels with high Mn, Al and C content [25–30]. Based on these technical merits, in this work, some strips in thickness of 2.5 mm were produced by a centrifugal casting equipment under a near-rapid solidification condition. These strips were evaluated for microstructure and mechanical properties.

## 2. Experimental

Composition of studied material is expressed as Fe–28Mn–9Al-1.2C-$x$Cr ($x$ = 0, 3, 6, 9) (wt.%). An alloy ingot was made from high purity Fe (99.99%), Mn (99.9%), Al (99.9%), C (99.8%), and Cr (99.9%). The alloy ingot was reheated to melt by an induction coil, and then liquid metal was poured into a copper mold at a rotating speed of 600 r/min in Ar atmosphere, with calculated maximum cooling rate at about $5 \times 10^3$ K/s [31]. The obtained strip size was (60–80) mm × 60 mm × 2.5 mm. Some near-rapidly solidified strips were heat-treated in a vacuum electric resistance furnace to compare the properties. The actual composition of strips was detected by an Inductively Coupled Plasma mass spectrometry (ICP) as Table 1. The measured average density of the 0Cr, 3Cr, 6Cr, and 9Cr strips was 6.49 g/cm$^3$, 6.45 g/cm$^3$, 6.46 g/cm$^3$, and 6.43 g/cm$^3$, respectively, which was performed by an SJ-600G Electronic Water Displacement Density Analyzer.

The microstructure was characterized by an optical microscope (OM, Zeiss Axio imager A2m) and a transmission electron microscope (TEM, JEM-2100F, operated at a voltage of 200 kV). The constituent phases were identified by an X-ray diffractometer (XRD, D/Max-2200, Cu target, operated at 40 kV and 40 mA) with a scan rate of 4° per min. Tensile specimens were made in a gauge size of 20 mm × 4 mm × 1 mm, and tensile tests were conducted at a strain rate of $1 \times 10^{-3}$ s$^{-1}$ by a MTS Criterion Model 44 mechanical property tester with a 10 kN capacity, each test data was supported by more than three tensile specimens.

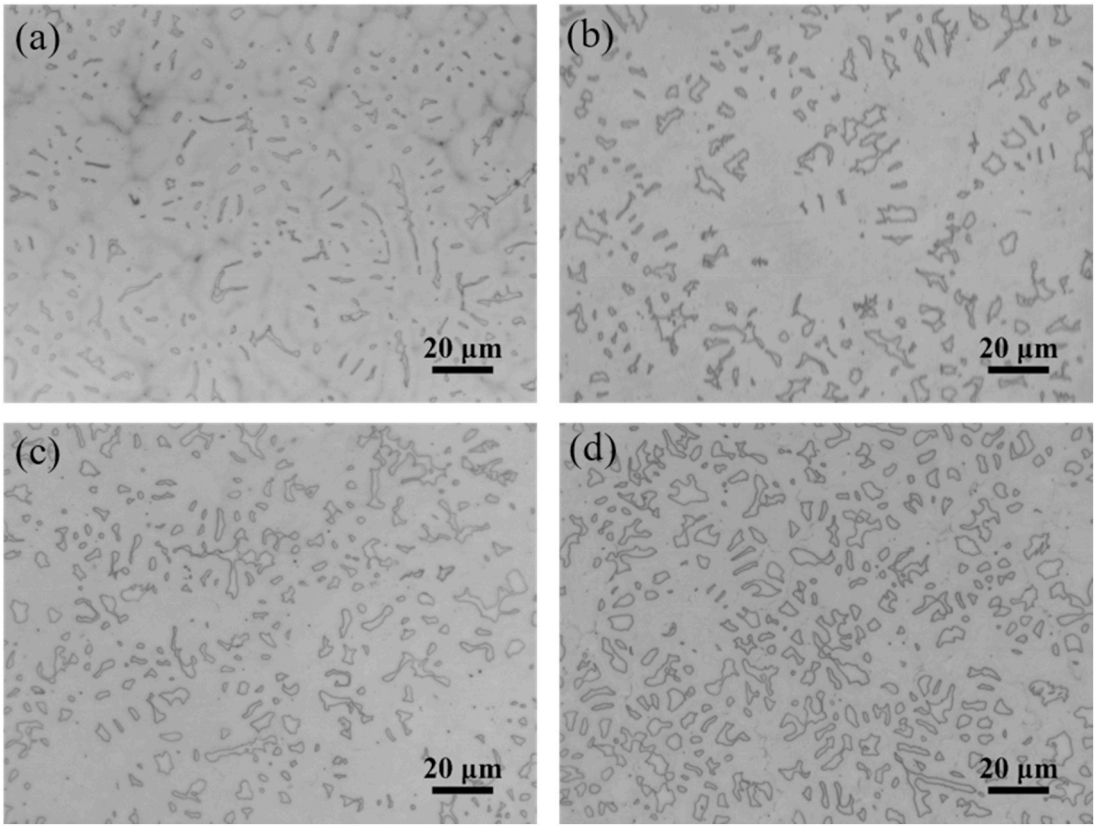



**Fig. 2.** OM of as-cast Fe–28Mn–9Al-1.2C-$x$Cr low-density strips: (a) 0Cr, (b) 3Cr, (c) 6Cr, (d) 9Cr.

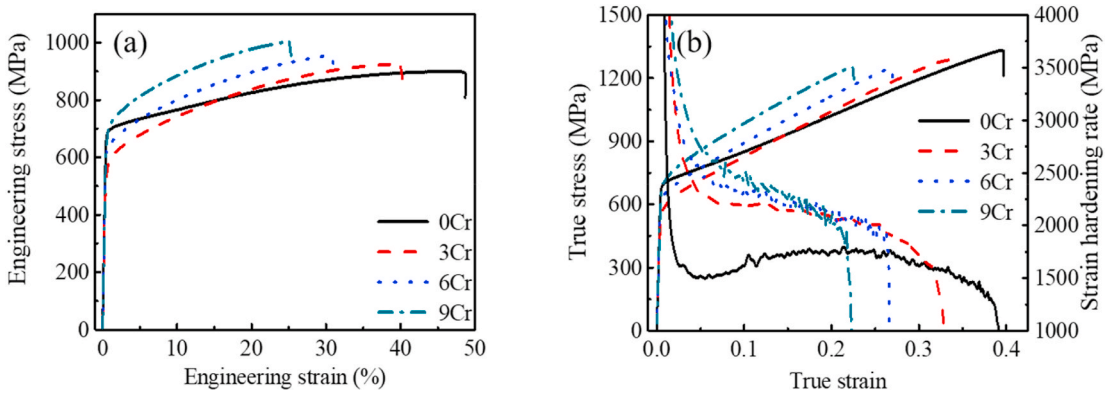



**Fig. 3.** Tensile property of as-cast Fe–28Mn–9Al-1.2C-*x*Cr: (a) engineering stress strain curves, (b) true stress strain and strain hardening rate curves.

**Table 2**
Some specific values of tensile experiment of as-cast and annealed (at 550 °C for 5 h) Fe–28Mn–9Al-1.2C-*x*Cr strips.

| | As-cast | | | Annealed | | | Increment of YS after annealing (MPa) | Increment of TE after annealing (%) |
|---|---|---|---|---|---|---|---|---|
| | YS(MPa) | UTS (MPa) | TE (%) | YS (MPa) | UTS (MPa) | TE (%) | | |
| 0Cr | 724 ± 16 | 924 ± 24 | 40.8 ± 8 | 882 ± 6 | 934 ± 4 | 30.1 ± 2 | + 158 | - 10.7 |
| 3Cr | 617 ± 4 | 896 ± 28 | 33.2 ± 7 | 768 ± 9 | 915 ± 23 | 37.4 ± 5 | + 151 | + 4.2 |
| 6Cr | 652 ± 5 | 953 ± 9 | 31.6 ± 1 | 751 ± 2 | 965 ± 2 | 44.3 ± 3 | + 101 | + 12.7 |
| 9Cr | 740 ± 8 | 979 ± 22 | 20.3 ± 5 | 805 ± 11 | 994 ± 21 | 22.8 ± 4 | + 65 | + 2.5 |

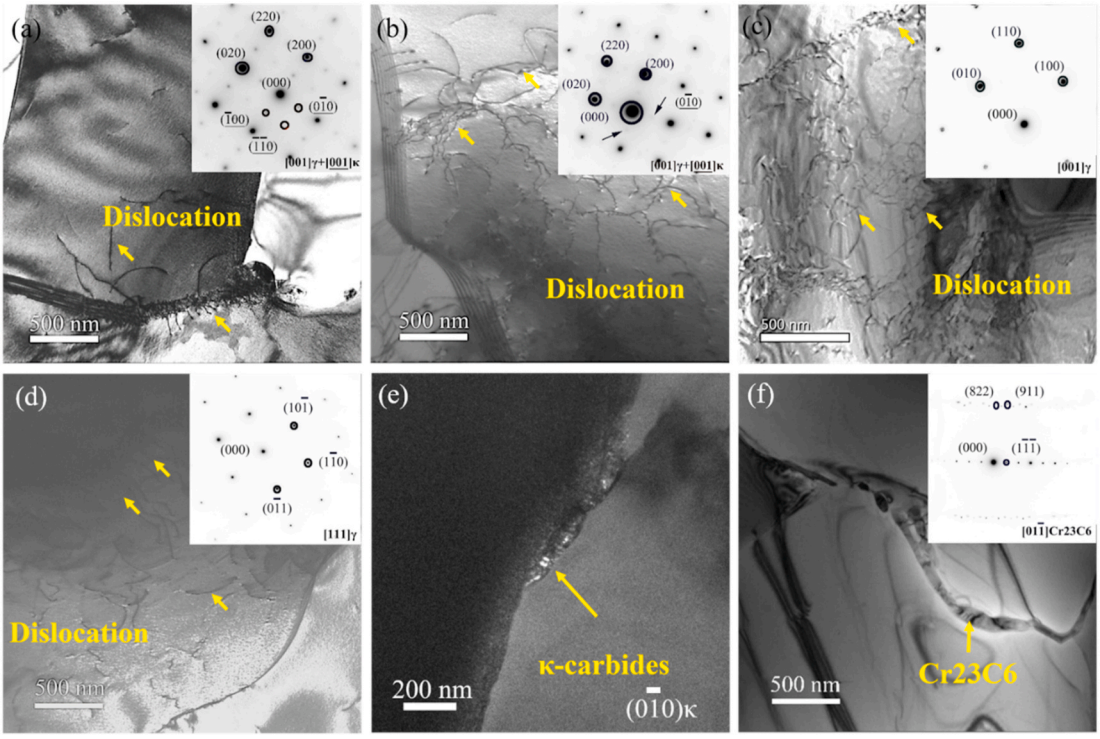



**Fig. 4.** TEM images of austenite in Fe–28Mn–9Al-1.2C-*x*Cr near-rapid solidified strips: (a) BF image of 0Cr strip and corresponding SADP, (b) BF image of 3Cr strip and corresponding SADP, (c) BF image of 6Cr strip and corresponding SADP, (d) BF image of 9Cr strip and corresponding SADP, (e) DF image of 0Cr strip using $(010)_{\kappa}$, (f) BF image of 9Cr strip with $Cr_{23}C_6$ and corresponding SADP.

## 3. Experimental results

### 3.1. Microstructure and mechanical properties of as-cast strips

XRD results indicate that all the Fe–28Mn–9Al-1.2C-*x*Cr strips are composed of austenite (denoted as γ) and ferrite (denoted as δ), and austenite is the major matrix phase, as shown in Fig. 1. With an increase in Cr content, the ferrite phase (110) diffraction ($2\theta = 44.1°$) intensity increases, which means that addition of Cr promotes the formation of ferrite phase. According to XRD results by JADE, the austenite lattice constants are 3.70 Å, 3.56 Å, 3.56 Å and 3.55 Å for 0Cr, 3Cr, 6Cr, and 9Cr strips, respectively. The metallographic photos indicate that ferrite

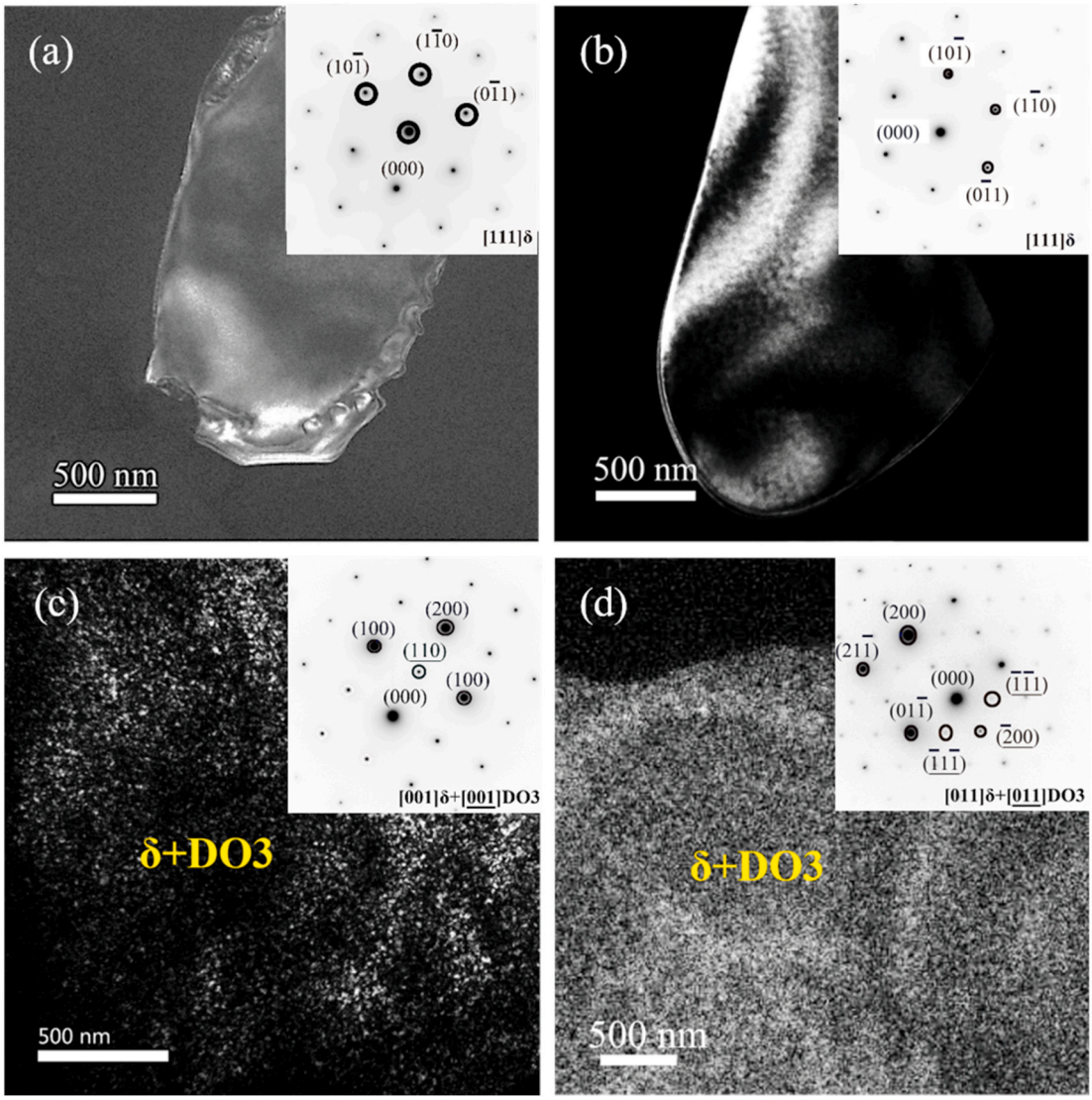



**Fig. 5.** TEM images of ferrite in Fe–28Mn–9Al-1.2C-*x*Cr strips: (a) DF image of 0Cr strip using $(1\bar{1}0)_\delta$ and corresponding SADP; (b) DF image of 3Cr strip using $(1\bar{1}0)_\delta$ and corresponding SADP; (c) DF image of 6Cr strip using 200">$(200)_{DO3}$ and corresponding SADP; (d) DF image of 9Cr strip using $(200)_{DO3}$ and corresponding SADP.

uniformly distributes in the austenite matrix, as shown in Fig. 2. According to a quantitative measurement by Gatan Digital Micrograph, the ferrite fraction is 10.2 vol %, 11.4 vol%, 14.8 vol%, and 19.6 vol% for 0Cr, 3Cr, 6Cr, and 9Cr strips, respectively. Moreover, Cr addition changes the shape of ferrite from rodlike to block like gradually. It is found from Fig. 2 (a) that all ferrites are in the center of austenite grains, which means that δ-ferrite first nucleated and grew from melt during the near-rapid solidification. The experimental results agree with the solidification mode of this alloy: FA (ferrite-to-austenite) [2,4], i.e. L (liquid) → L+δ (high temperature ferrite) → L+δ+γ (austenite) → δ+γ.

Fig. 3 (a) shows the engineering stress-strain curves of as-cast Fe–28Mn–9Al-1.2C-*x*Cr strips. The results indicate that with an increase in Cr content, yield strength first decreases from 724 MPa to 617 MPa and then increases to 740 MPa, but ultimate tensile strength increases and total elongation decreases gradually. Fig. 3 (b) is the true stress-strain and strain hardening rate curves. It is found that Cr addition greatly enhances strain hardening rate in the steady hardening stage, and it gradually increases with higher Cr content, which the average rates are 1626 MPa, 2048 MPa, 2266 MPa and 2385 MPa, respectively, which has the similar phenomenon in Zhang' s result [17]. The Cr addition increases the ultimate tensile strength, though total elongation decreases gradually. The detailed value about tensile properties of this research are shown in Table 2, averaged over 3 measurements.

To find the origin for the changes in mechanical properties, TEM was carried out on as-cast strips. TEM photographs of Fe–28Mn–9Al-1.2C-*x*Cr strips are shown in Fig. 4. It is found that there are a few dislocations in the austenite of 0Cr strip, as shown in bright-field (BF) image in Fig. 4 (a). When 3 wt% and 6 wt% Cr are added to the strips, more dislocations appear in the austenite, as shown in Fig. 4 (b, c). The dislocation density in 9Cr strip is still very high, but slightly lower than that of 6Cr strip, as shown in Fig. 4 (d). Those indicate that addition of Cr increases dislocation density in austenite matrix of Fe–28Mn–9Al-1.2C strips. Selected-area diffraction patterns (SADPs) are inserted in Fig. 4.

There is a clear superlattice in austenite matrix of 0Cr strip, which is characterized as κ-carbides. But the superlattice in austenite diffraction pattern of 3Cr and 6 Cr strips is very weak, as shown in Fig. 4 (b) and **(c)**. No superlattice belonging to κ-carbides is observed in austenite diffraction pattern of 9Cr strip as shown in Fig. 4 (d). The sizes of κ-carbides in the austenite of these three strips are too small to identify, and κ-carbides are only visible at phase boundary between austenite and ferrite in dark-field (DF) image of 0Cr strip, as shown in Fig. 4 (e). Thus, addition of Cr suppresses the precipitation of κ-carbide as the results shown in some researches [17,18]. However, some $Cr_{23}C_6$ carbides are also observed at austenite grain boundary in 9Cr strip, which has a cubic

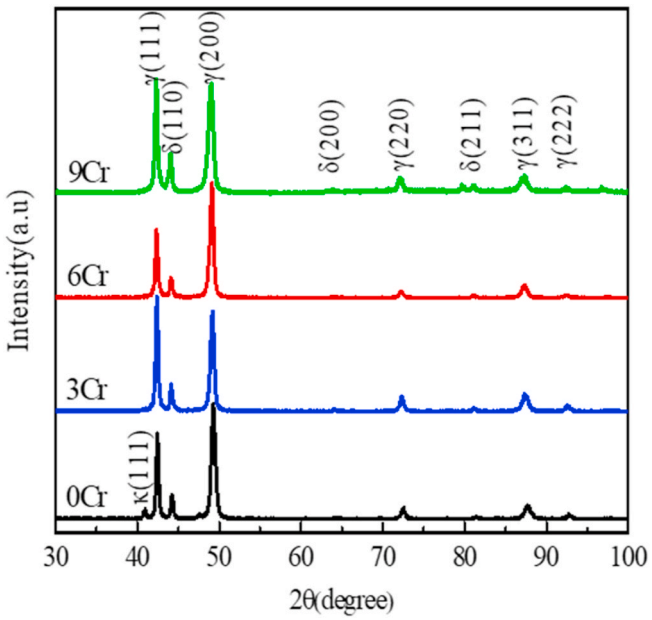



**Fig. 6.** X-ray diffraction patterns of Fe–28Mn–9Al-1.2C-*x*Cr strips after annealing at 550 °C for 5 h.

structure with a lattice constant of 1.054 nm, as shown in Fig. 4 (f). The $Cr_{23}C_6$ carbides are in granular or banded shapes, with a width of about 100 nm, which may lead to inter-granular cracking along coarsening carbides at grain boundary [23].

Only a few dislocations are occasionally observed in the ferrite in all strips. In the ferrite of 0Cr and 3Cr strips, as shown in Fig. 5 (a) and (b), no superlattice is observed, and the images exhibit almost clean features without evident secondary phases. In 6Cr and 9Cr strips, a superlattice belonging to DO3 phase in ferrite is observed, as shown in Fig. 5 (c) and (d), which is determined by the fact that (*hkl*) are all odd or all even, and the reflections belong to DO3 phase are similar with previous results [17]. According to DF images of ferrite, nano-scale DO3 phases clearly appear in ferrite of 6Cr and 9Cr strips.

As mentioned above, addition of Cr suppressed the formation of nano-scale κ-carbides while increased dislocation density. Reduction of nano-scale κ-carbides could lower down their yield strengths [14,17], while precipitation of $Cr_{23}C_6$ and increase in dislocation density increased their yield strengths. Because of the obstruction of κ-carbides to the dislocation glide and generation of the slip bands, reduced precipitation of the κ-carbides in Cr-containing strips results in a high strain hardening rate [17]. The experiment shows that, with the Cr addition, the yield strength decreases first until 6 wt% Cr then increases, and the yield strength of 9 wt% is even higher than that of 0Cr. Although yield strength and total elongation of 6Cr strip are lower than those of Cr-free strip, its tensile strength is higher than that of Cr-free strip due to a higher strain hardening rate. Glide plane softening effect of nano κ-carbides yields a lower straining hardening rate of as-cast Cr-free strip [4,17,32].

The reason why the as-cast Cr-containing strips have lower plasticity may come from the following factors. The obtained strips are almost free of casting defects due to higher feeding capacity of the melt supported by great centrifugal force; moreover, any present casting defects are randomly distributed. So casting defects may not be considered as a reason for the decrease of plasticity in Cr-containing strips. Increase of δ-ferrite fraction in Cr-containing strips can blocks the dislocation propagation due to more phase-interface areas, which may decrease the plasticity. Reduction in fraction and size of nano-scale κ-carbides could increase its plasticity [12,15,18], which is not the reason for the decrease in plasticity of 3Cr and 6Cr strips. The increased dislocation density in Cr-containing strips and formation of $Cr_{23}C_6$ carbides in 9Cr

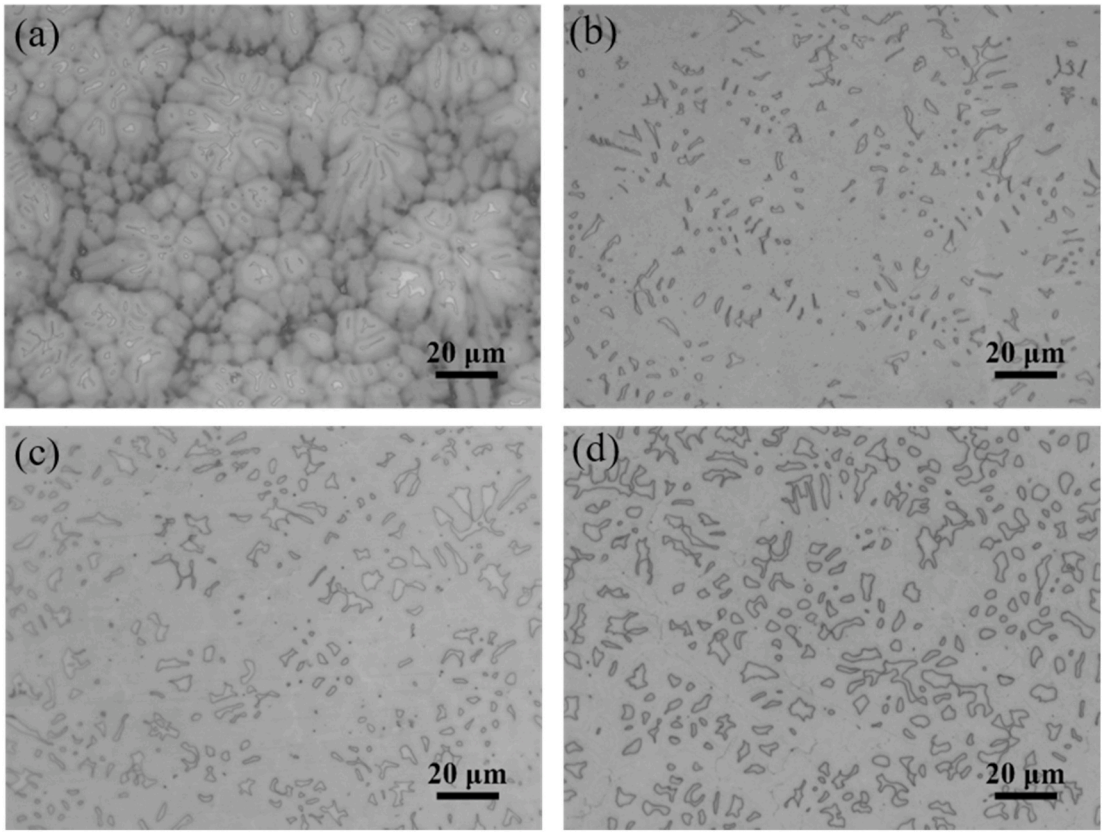



**Fig. 7.** SEM pictures of Fe–28Mn–9Al-1.2C-*x*Cr low-density steel strips after annealing at 550 °C for 5 h: (a) 0Cr, (b) 3Cr, (c) 6Cr, (d) 9Cr.

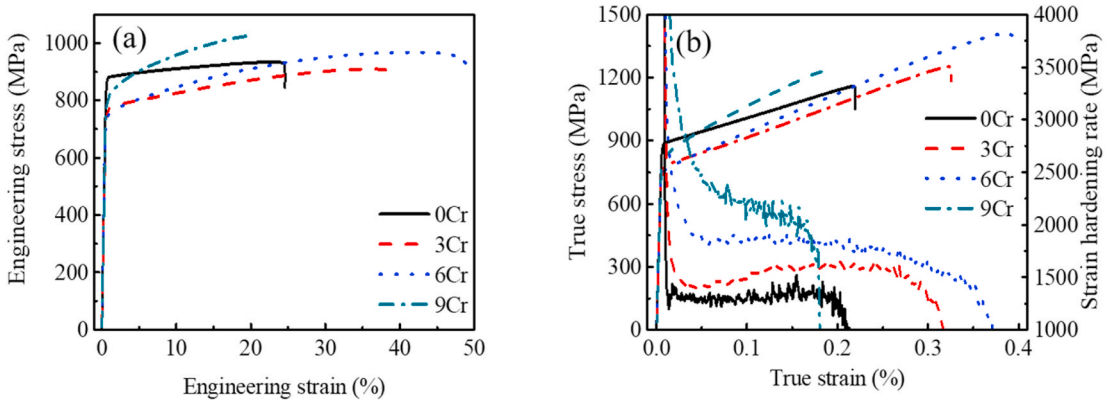



**Fig. 8.** Tensile property of Fe–28Mn–9Al-1.2C-*x*Cr annealed at 550 °C for 5 h: (a) Engineering stress strain curves, (b) True stress strain and strain hardening rate curves.

are the likely factors decreasing the plasticity.

The dislocation density in Cr-containing strips may vary in the following steps. Under the same solidification condition, the residual stress of each strips caused by rapid cooling is similar [30]. Moreover, reduction of fraction/size of nano-scale κ-carbides increases the formation ability of dislocations under the same thermal stress caused by rapid cooling rate [32–36]. Higher strain hardening rate of Cr-containing strips also confirms higher formation ability of dislocations. Therefore, the dislocation density of Cr-containing strips is higher under the same solidification condition for 3Cr and 6Cr strips. The addition of 9 wt. % Cr further improves the solid solubility of C and inhibits the precipitation of κ-carbides, while the precipitation of Cr23C6 in the 9Cr sample reduce the formation of dislocations [33,36], leading to reduction of dislocations in austenite.

### *3.2. Microstructures and mechanical properties of the strips after annealing*

The constituent phase of Fe–28Mn–9Al-1.2C-*x*Cr strips after annealing at 550 °C for 5 h are still austenite and ferrite, as shown in Fig. 6 and Fig. 7. The XRD result in Fig. 6 displays that the diffraction peak (111) of κ-carbides at 41.4° obviously appears in annealed 0Cr strip, suggesting that annealing at 550 °C for 5 h promotes the precipitation of κ-carbides. However, this diffraction peak is still not observed in Cr-containing annealed strips, so Cr addition also suppresses the precipitation of κ-carbides even after annealing. As shown in Fig. 7 (a), the grain boundaries and dendritic structure of annealed 0Cr strip are clearly observed, while the grain boundary could not be observed in Cr-containing strips under the same conditions, indicating the addition of Cr improves corrosion resistance of steel strips as reported in the research of Tsay and Tuan [20,22,23]. Compared with the as-cast strips,

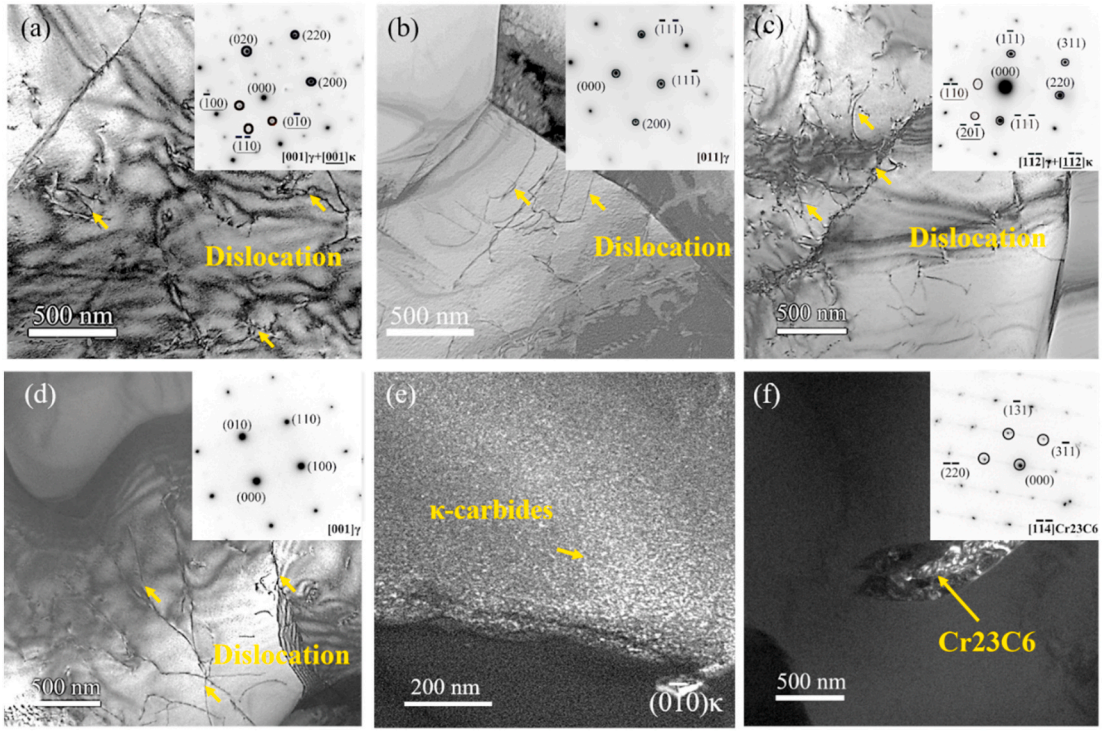



**Fig. 9.** TEM images of austenite in Fe–28Mn–9Al-1.2C-*x*Cr annealing at 550 °C for 5 h: (a) BF image of 0Cr strip and corresponding SADP, (b) BF image of 3Cr strip and corresponding SADP, (c) BF image of 6Cr strip and corresponding SADP, (d) BF image of 9Cr strip and corresponding SADP, (e) DF image of 0Cr strip along $(010)_{\kappa}$, (f) DF image of 9Cr strip along$(\bar{1}44)_{Cr23C6}$ and corresponding SADP.

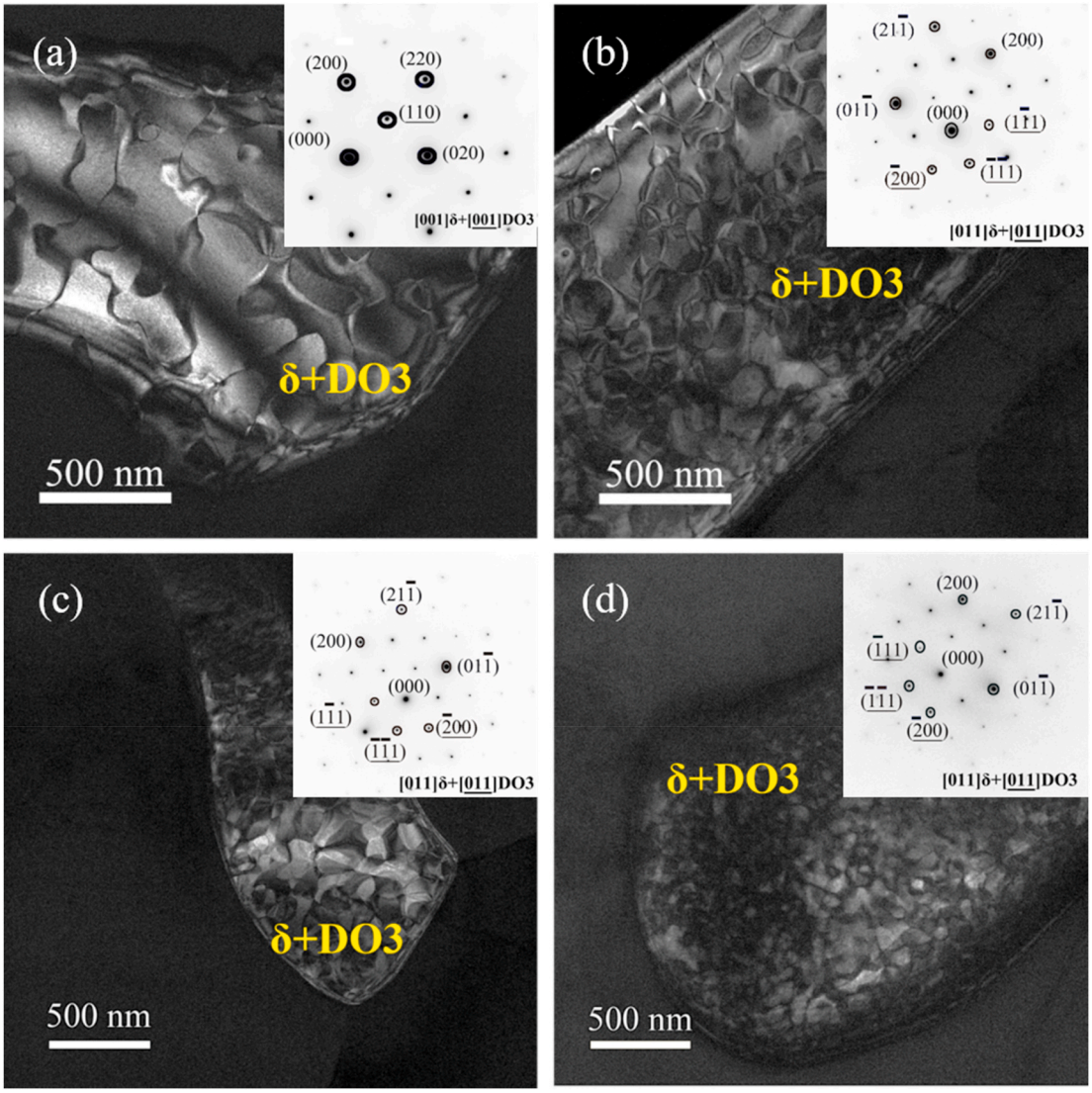



**Fig. 10.** TEM images of ferrite in Fe–28Mn–9Al-1.2C-*x*Cr annealing at 550 °C for 5 h: (a) DF image of 0Cr strip along$(1\bar{1}0)_\delta$ and corresponding SADP, (b) DF image of 3Cr strip along $(1\bar{1}0)_\delta$ and corresponding SADP, (c) DF image of 6Cr strip along $(200)_{DO3}$ and corresponding SADP, (d) DF image of 9Cr strip along $(200)_{DO3}$ and corresponding SADP.

the size and fraction of ferrite in annealed strips almost remain the same.

Fig. 8 (a) is the engineering stress strain curves of annealed Fe–28Mn–9Al-1.2C-*x*Cr strips. It can be found that annealing at 550 °C for 5 h significantly enhances yield strength of all the strips, while Cr addition reduces the increment of yield strength. After annealing, the total elongation of 0Cr strip obviously reduces but that of 6Cr strip clearly increases. An observed unusual phenomenon is that with an increase of Cr content, the total elongation of as-cast strips gradually decreases, while that of annealed strips gradually increases except of the strip containing 9 wt% Cr. The detailed value about tensile properties of the annealed strips are shown in Table 1 as mentioned above.

The true stress-strain and strain hardening rate curves of annealed strips are shown in Fig. 8 (b), which also indicates that the strain hardening rate is higher with an increase of Cr content in the annealed strips. Compared with as-cast strips, the strain hardening rate of all annealed strips except 9Cr strip clearly decreases. The reduction in the strain hardening rate of annealed 3Cr strip is the largest about 500 MPa.

To correlate the microstructure with the mechanical properties of annealed strips, TEM was conducted on annealed Fe–28Mn–9Al-1.2C-*x*Cr strips, and the results of TEM are shown in Fig. 9 for austenite and Fig. 10 for ferrite.

After annealing at 550 °C for 5 h, there are still clear superlattices of κ-carbides in austenite matrix of 0Cr strip, and its intensity increases obviously as shown in Fig. 9 (a). The superlattices of κ-carbides in SADP of annealed 3Cr and 6Cr strips are still very weak, as shown in Fig. 9 (b, c). No superlattice of κ-carbides in SADP is found, and only $Cr_{23}C_6$ carbides with a width of about 200 nm are observed in annealed 9Cr strip, as shown in Fig. 9 (f). The dislocation density in austenite matrix of annealed Cr-containing strips is obviously reduced, compared with those in as-cast Cr-containing strips, while it is still higher than that of annealed 0Cr strip. After annealing at 550 °C for 5 h, the superlattice belonging to DO3 phase appears in all annealed strips, including 0Cr and 3Cr, as shown in Fig. 10.

TEM and XRD results indicate that the annealing treatment increases κ-carbides volume fraction and decreases dislocation density. Thus, the increase of yield strength should be attributed to the increase of κ-carbides in the strips. The increase of κ-carbides can reduce plasticity, so the total elongation of Cr-free strip clearly decreases due to the presence of κ-carbide precipitation. While less precipitation of κ-carbides and reduced dislocation density are presented in 3Cr and 6Cr strips, yielding small increments in yield strength and elongation. The precipitation of nano-scale κ-carbides leads to lower strain hardening rate due to glide

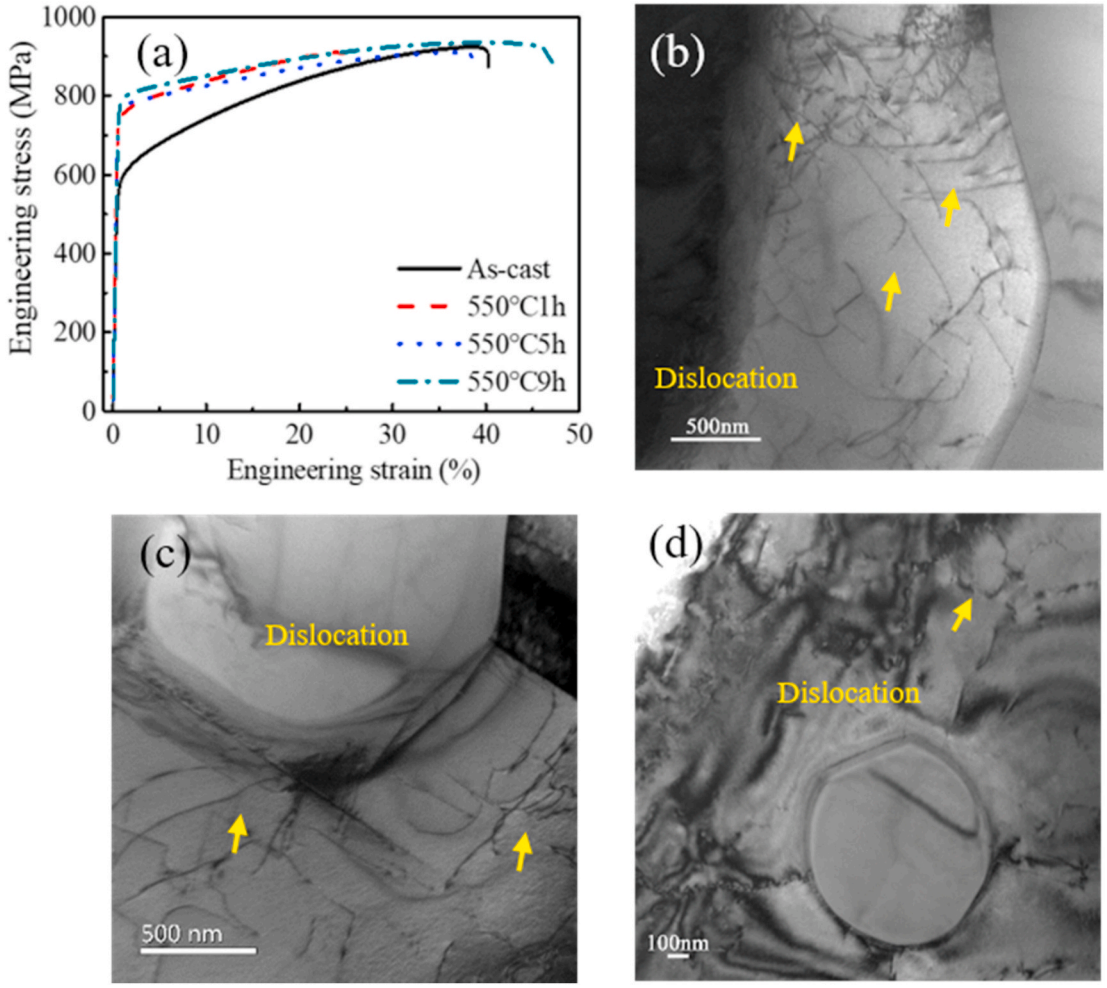



**Fig. 11.** Engineering stress strain curves and TEM images of austenite in Fe–28Mn–9Al-1.2C–3Cr annealing at 550 °C for different time: (a) Engineering stress strain curves; (b) BF image of 1 h specimens; (c) BF image of 5 h specimens; (d) BF image of 9 h specimens.

plane softening effect [4,17,32]. The content of κ-carbides is also confirmed by the strain hardening rate of annealed strips. The appearance of $Cr_{23}C_6$ carbide particles at grain boundary is the main reason for the poor ductility and high strain hardening rate of as-cast and annealed 9Cr strips.

To confirm the reason for improvement in plasticity of annealed Cr-containing strips, the mechanical properties and microstructures of 3Cr strips were examined after different annealing time at 550 °C. As shown in Fig. 11 (a), the plasticity of 3Cr strip increases gradually with an increase of annealing time, but the yield strength do not change significantly after 1 h annealing. By observing TEM images, it is found that the dislocation density gradually decreases with increasing the annealing time, as shown in Fig. 11(b), (c) and **(d)**, respectively. The increase of yield strength is resulted from precipitation of nano-scale κ-carbides. The evolution of properties and microstructures as increasing of annealing time at 550 °C indicates that κ-carbides are mainly precipitated in the first hour of annealing treatment, while the recovery needs longer time. So to improve the plasticity, the annealing time should be prolonged to reduce dislocation density at this annealing temperature, while longer annealing time may coarsen κ-carbides [34]. The anneal time should be further optimized to achieve a balance between the coarsening of κ-carbides and reduction of dislocation density.

## 4. Discussion

As mentioned above, Cr addition can reduce the precipitation of κ-carbides in as-cast and annealed strips, which is consistent with the previous results [17]. The reason is that addition of Cr can expand the solid solubility of Al and C in the Fe–Mn–Al–C steel [35,36], and reduce the driving force of the formation of κ-carbide. Therefore, the higher the Cr content is, the less the κ-carbide precipitates. In this work, it is found that $Cr_{23}C_6$ carbides are generated when the Cr content reaches to 9 wt %. On the other hand, there are some dislocations in near-rapidly solidified strips, because rapid cooling rate inevitably leads to thermal stress during near-rapid solidification process, which results in the formation of dislocations [27,28,37], and more dislocations form in Cr-containing as-cast strips [38–41].

In this work, slip bands in austenite are observed during deformation of the strips, thus we mainly examin the slip bands of 0Cr and 6Cr strips after tensile fracture. Fig. 12(a) and (c) are TEM images of the microstructure in as-cast 0Cr and 6Cr strips after tensile fracture. Average spacing between slip bands is ~90 nm and ~80 nm in as-cast 0Cr and 6Cr strips, respectively. For Fe–Mn–Al–C low-density steel, average spacing between slip bands has a dependence to deformation behavior. The narrowed average spacing between slip bands usually suggests higher strain hardening [32,42,43], which is in a good agreement with our results. The narrowed average spacing between slip bands usually causes higher elongation after tensile fracture, which is not consistent with our work, because 0Cr strip with larger average spacing has higher elongation than that of 6Cr strip. From the TEM images shown in Fig. 4, there are more dislocations in austenite of as-cast 6Cr strip before stretching than that of 0Cr strip, which should be one of the reasons for

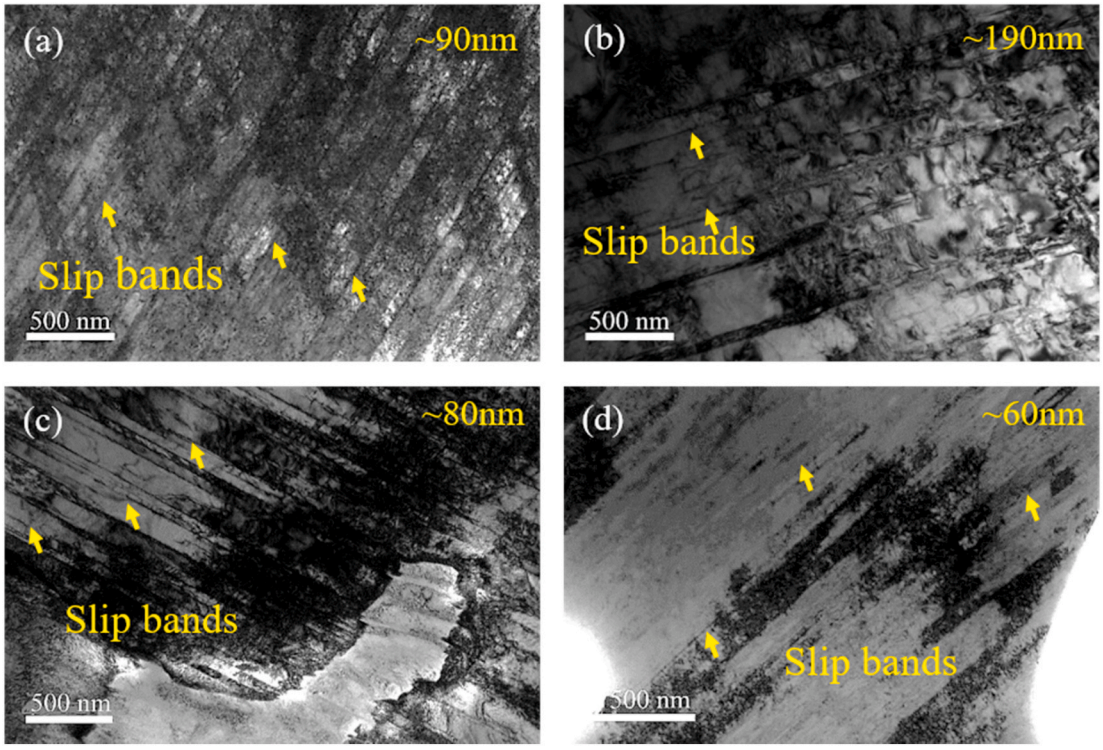



**Fig. 12.** TEM images of 0Cr and 6Cr steel strips after tensile fracture: (a) as-cast 0Cr steel strip, (b) annealed 0Cr steel strip, (c) as-cast 6Cr steel strip, (d) annealed 6Cr steel strip.

low elongation of 6Cr strip. On the other hand, if a κ-carbide is sheared by a dislocation, the following dislocations can easily cut through the sheared κ-carbides rather than non-sheared κ-carbides [4,44]. As a result, slip spacing is relatively wide by the localization of the shearing due to higher obstructive of larger κ-carbides for dislocations in 0Cr strip. In the as-cast 6Cr strip, because of the smaller and less κ-carbides which are relatively easy to cut through, multi-directional shearing occur, leading to a narrow slip band spacing.

Fig. 12(b) and (d) are TEM images of annealed 0Cr and 6Cr strips after tensile fracture. After tensile fracture, average spacing of 0Cr strip increases to ~190 nm, but that of 6Cr strip decreases to ~60 nm compared with as-cast strips, which is also evidenced by increase in elongation of 6Cr strip and decrease in elongation of 0Cr strip. It is an interesting phenomenon that although annealing promotes precipitation of the κ-carbides in 0Cr and 6Cr strips, which is confirmed by changing their yield strength and TEM images, an opposite change occurs in average spacing in annealed 0Cr and 6Cr strips. In the annealed 0Cr strip, more κ-carbides are present with an average diameter of 14 nm precipitate in austenite, as shown in Fig. 9, and thus these coarse carbides are barriers for dislocations because dislocations can only pass through those with diameter being smaller than 10 nm [32,45], while the average diameter of κ-carbides is 8 nm in as-cast 0Cr strip.

Therefore, less slip bands are formed by shear mechanism during stretching process, and total elongation significantly decreases. While in annealed 6Cr strip, less κ-carbides are present with smaller diameter, which can be easily passed through by dislocations. Thus, the precipitated κ-carbides of 6Cr strips after annealing slightly affect total elongation. Moreover, thermal stress and dislocations existed in as-cast 6Cr strip partially disappear after annealing, allowing more slip bands to generate during stretching process to increase its elongation.

A statistical measurement indicates that average widths of slip bands in as-cast 0Cr and 6Cr strips are 13.4 nm and 12.5 nm, and the average widths of slip bands in annealed 0Cr and 6Cr strips are 20.5 nm and 7.9 nm, respectively, after tensile fracture as shown in Fig. 12. These indicate that average width of slip bands is inversely correlated with spacing

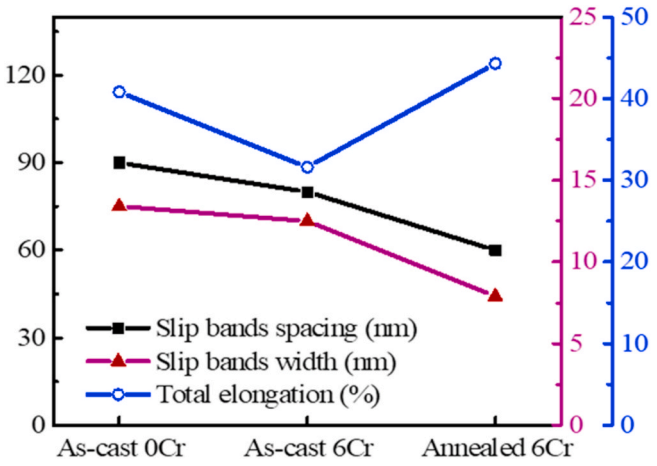



**Fig. 13.** The slip band spacing and width after tensile fracture of 0Cr and 6Cr strips with different status.

between slip bands after tensile fracture. Finer slip bands and smaller average spacing after tensile fracture are usually corresponding to high ductility, and this phenomenon is termed as dynamic slip band refinement (DSBR) [35,43,46]. However, although κ-carbides in as-cast 6Cr strip have the lowest content and the smallest size among three strips, the total elongation after tensile fracture is lowest (as shown in Fig. 13) due to its high dislocation density before stretching, which interferes the relationship between slip bands and ductility. It can be concluded that the dislocations play an important role in ductility of this austenitic low-density steel, besides formation ability of slip bands during deformation process.

## 5. Conclusions

In present work, Fe–28Mn–9Al-1.2C-*x*Cr low-density steel strips were prepared by a near-net shape method under near-rapid solidification condition. The effects of Cr addition on microstructures and mechanical properties were investigated. The following conclusions were made.

(1) Cr addition increased the ferrite fraction of Fe–28Mn–9Al-1.2C-*x*Cr strips prepared under near-rapid solidification conditions, but reduced the precipitation of κ-carbides in austenite. $Cr_{23}C_6$ carbides precipitated in 9Cr strip, instead of κ-carbides. Moreover, more dislocations appeared in Cr-containing strips.
(2) With an increase of Cr content, yield strength of as-cast Fe–28Mn–9Al-1.2C-*x*Cr strips decreased first due to reduction of κ-carbides, and then increased gradually because of the increase in dislocation density and the formation of $Cr_{23}C_6$ carbides. The increase in dislocation density and formation of $Cr_{23}C_6$ carbides reduced the ductility of Cr-containing strips gradually.
(3) Annealing at 550 °C for 5 h promoted the precipitation of κ-carbides, but content and size of κ-carbides in Cr-containing strips were still lower than those of Cr-free strip. The yield strength of all strips significantly increased, but an opposite change in ductility occurred between Cr-free and Cr-containing strips. The total elongation of annealed 6Cr strip was better than that of annealed Cr-free strip. The formation of $Cr_{23}C_6$ carbides caused the lowest ductility of as-cast and annealed 9Cr strips.
(4) Analysis of TEM images in 0Cr and 6Cr strips before and after tensile fracture suggested that initial dislocation density also had an important influence on elongation, besides the formation ability of slip bands during deformation process.

## CRediT authorship contribution statement

**Yuxiang Liu:** Conceptualization, Data curation, Writing – original draft, Investigation. **Mingxiang Liu:** Visualization. **Jianlei Zhang:** Visualization. **Wei He:** Investigation, Data curation, Visualization. **Zhiping Luo:** Conceptualization, Writing – review & editing. **Changjiang Song:** Conceptualization, Methodology, Formal analysis, Project administration, Funding acquisition, Writing – review & editing. **Qijie Zhai:** Supervision, Resources.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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